Boxuan Wang

Design Engineering Student

Product Design | AI | Robotics | jw4424@ic.ac.uk

PROFILE

Design Engineering student at Imperial College London with a broad technical foundation. Combines product planning, prototyping and programming to turn ideas into working prototypes and tools.

EDUCATION

Imperial College London 2024 - 2028

Design Engineering | Expected completion 2028

Relevant study: Mechanical Engineering, FEA, Materials and Manufacturing, Physical Computing, Intelligent Systems.

The King's School, Canterbury 2019 - 2024

King's Scholar; Vice Captain of Fencing.

WORK EXPERIENCE

Li Auto, *Humanoid Development Intern* July - September 2026

Humanoid development for commercial applications, including manufacturing.

- Designed connector parts and structural supports in SolidWorks, supported by Ansys analysis.
- Redesigned upper-torso load paths as part of the team, reducing torso weight by **18%** and meeting fatigue requirements in Ansys simulations with a safety factor of 3.
- Explored AI agents and MCP tools for 3D-printable validation structures; developed a reusable agent skill for temporary prototypes.
- Collected human-motion data with wearable trackers, VR headsets and robot arms mirroring operators, for tasks including parts organisation.

Elibot, *Cooperative Robot Intern* July - August 2024

- Built a Python system to log incoming, produced and shipped robots.
- Created joint-specific fatigue tests, logged faults and examined angular differences during braking in harmonic-drive joints.
- Contributed to app/web interfaces for remote robot control and webpage UX.
- Translated robot manuals into English and contributed to their layout and visual design.

SELECTED PROJECTS

Street Pulse Academic team project

- Conducted most user research; developed the tessellation and exterior design.
- Prototyped the electronics and carried out assembly and testing; software was developed by teammates.

Autonomous Football Robot 2025

- Contributed to team ball-tracking model training and control-code development using OpenMV and Edge Impulse on an existing robot platform.

RE-seat 2025

- Explored materials, structural analysis and lifecycle impact using Eco Audit for a seat concept.

SKILLS

Product design. Product planning, concept development, UX/interaction design; Figma, Miro.

Prototyping. SolidWorks, Fusion; CAD, design for additive manufacturing, prototype assembly.

Programming and AI. Python, C++, MATLAB; OpenMV, Edge Impulse; AI agents, MCP tools.

Communication. InDesign, Overleaf; presentations and technical documentation.

Supporting analysis. Experience with Ansys Mechanical, Fluent and Discovery; FEA/CFD and joint testing.

LANGUAGES

Chinese (Native) | **English** (Fluent) | **French** (Basic)